An Explainable Deep Learning System for Cyberattack Detection in Internet of Energy Networks

Makhduma F. Saiyed, Irfan Al-Anbagi, and M. Shamim Hossain

Abstract

The integration of smart devices and real-time analytics in the Internet of Energy (IoE) has enabled intelligent, data-driven energy systems but also introduced cybersecurity concerns. Addressing these concerns requires detection mechanisms that are not only accurate but also interpretable. This article proposes an explainable deep learning system, named E-SHIELD, designed to detect various cyberattacks in IoE networks. The system uses a hybrid Convolutional Neural Network—Long-Short-Term Memory (CNN-LSTM) architecture to learn both spatial correlations and temporal dependencies from network traffic data. The architecture is optimized for resource-constrained environments by using compact filters, dropout regularization, and low learning rates. E-SHIELD is evaluated using two benchmark datasets, WUSTL-IIoT-2021 and CICIoT-2023, which contain diverse real-world attack scenarios such as Denial of Service (DoS), Distributed DDoS, Mirai variants, and fuzzing attacks. To ensure transparency, SHAP (SHapley Additive exPlanations) is integrated to provide both global and local interpretability of the model's predictions. A user dashboard has been developed to support human-in-the-loop monitoring, allowing users to analyze predictions, feature importance, and risk levels in real-time. Experimental results show that the E-SHIELD system achieves high accuracy compared to existing CNN, LSTM, and ensemble-based models. By combining robust detection capabilities with interpretable AI, E-SHIELD supports secure and trustworthy deployment in mission-critical energy systems.

Introduction

The Internet of Things (IoT) has brought a major transformation to traditional energy infrastructures by enabling interconnected environments where real-time monitoring, predictive analytics, and remote control are standard features. These advances allow for optimized energy distribution, early fault detection, and better user engagement. Building on these capabilities, the Internet of Energy (IoE) extends the concept of IoT by incorporating a variety of renewable and non-renewable energy sources into a unified digital framework that is closely integrated with smart grid technologies [1]. The IoE network links generation, transmission, distribution, and usage so they can share data. It uses smart tools like sensors and meters to run more efficiently, save money, and support a cleaner and greener energy future.

Security risks increase in proportion to the size and complexity of the IoE network. These networks use many small devices with limited resources that rely on different communication methods. Since many of these methods were not built with strong security, attackers can exploit weaknesses in authentication, firmware, or communication channels to compromise the availability, confidentiality, and integrity of the system [2]. There are many real examples where attackers have targeted energy systems [3], [4]. One case is the Triton malware, which attacked Schneider Electric's safety systems at a petrochemical plant in the Middle East. The malware tried to turn off the safety controls that protect people from accidents. In another case, Ukrainian attackers used BlackEnergy and Industroyer malware to hit the power grid. This caused blackouts that left thousands of people without electricity and disrupted important services. These incidents show that cyberattacks on energy systems can impact not only machines but also entire communities. The effect of such attacks can last for a long time [5].

In IoE networks, different types of cyberattacks are common. These include Denial-of-Service (DoS), Distributed Denial-of-Service (DDoS), False Data Injection (FDI), and fuzzing attacks. DoS and DDoS attacks overwhelm system resources, while FDI attacks manipulate sensor readings. Fuzz attacks exploit protocol vulnerabilities and firmware manipulation that alter device behaviour [6]. These attacks often take advantage of limited memory, slow communication, or misconfiguration of the network. Since there are numerous types of connected devices, it becomes difficult to apply the same security measures throughout the system. Many traditional security methods do not work well for IoE. They often require more computing power than edge devices can handle, or they are too slow to stop threats in real time [7].

To meet these challenges, researchers use Artificial Intelligence (AI) for the development of

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efficient and adaptive intrusion detection systems [8], [9], [10]. AI models, especially Deep Learning (DL) approaches, are good at finding patterns in large and complex data, helping them to detect new or changing types of threat effectively [8]. These models can also help with real-time detection and response, where even minor delays can cause problems [9], [10]. However, AI models have their own disadvantages. DL models operate like black boxes and do not reveal how they make decisions, which can make them difficult to trust. When AI systems make decisions without showing how they work, it can be difficult for grid operators, regulators, and users to fully trust or accept them, especially in critical energy systems where accountability is important [11]. To address this issue, explainable AI, or XAI, has become an integral part of AI-based detection systems. XAI helps by providing precise and easy-to-understand explanations of how the model makes its decisions. By highlighting which input features influence predictions, XAI helps system operators verify decisions, understand risks, and make informed responses [11]. This is especially important in IoE networks, where operators must review decisions quickly during emergencies. However, despite recent progress, existing methods still have important limitations. Some achieve high accuracy but lack interpretability, while others provide explanations but are too complex or slow for practical use of IoE.

To address these limitations, this article proposes an explainable DL system, namely E-SHIELD, that uses a hybrid CNN-LSTM architecture to detect cyberattacks in IoE networks. The E-SHIELD system detects various types of attack, including DoS, DDoS, fuzzing, and Mirai variants. The E-SHIELD system also provides interpretability through a SHAP-based explanation. Using data sets such as WUSTL-IIoT-2021 [12] and CICIoT-2023 [13], the system captures real-world attack scenarios and evaluates the performance of hybrid DL models. A user-friendly dashboard allows network professionals to visualize predictions, analyze the importance of features, and respond effectively to incidents. This work aims to close the gap between advanced AI-based detection and the need for clear and understandable results in critical infrastructure. The novelty of E-SHIELD lies in its ability to combine accurate threat detection with a SHAP-based explanation that gives the reason behind each decision. This helps bridge the gap between AI and the need for transparency in protecting critical infrastructure, such as energy systems.

The remainder of this article is organized as follows. The section "Architectural Layers and Data Flow in Internet of Energy Systems" presents the architectural layers. The section "Methodology" outlines the methodology used to design the E-SHIELD system. The section "Result Analysis" discusses the results. Finally, the section "Conclusion" concludes the paper and highlights directions for future work.

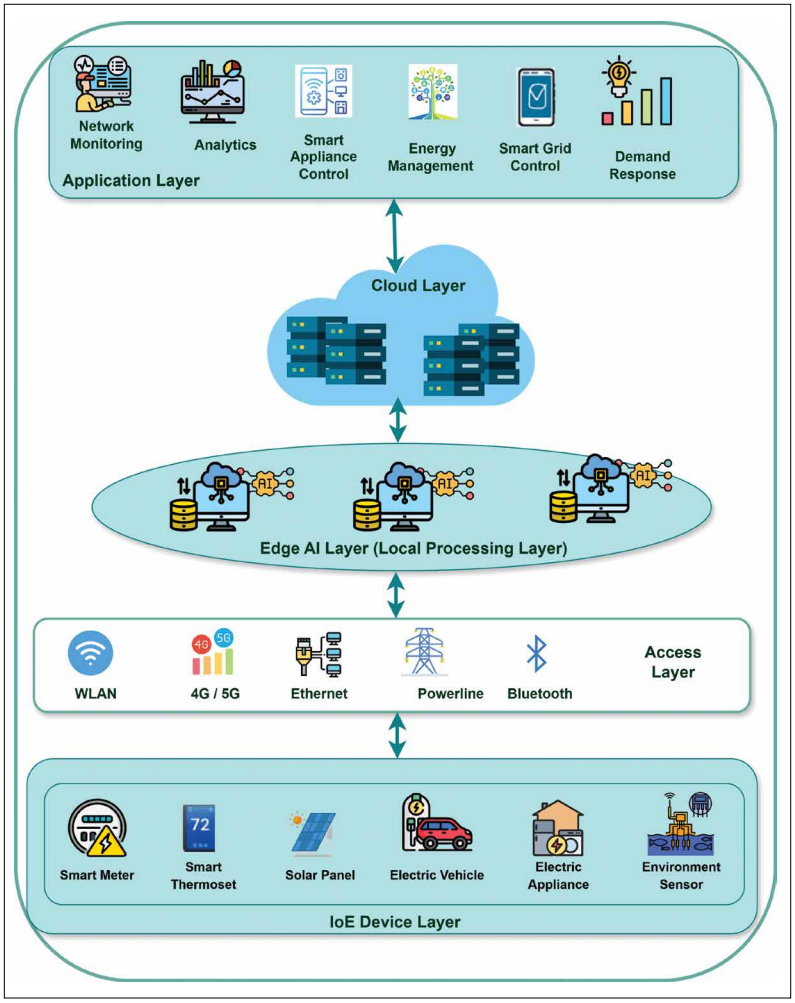


FIGURE 1. Layered Architecture and Data Flow in Internet of Energy Systems.

Architectural Layers and Data Flow in Internet of Energy Systems

The IoE ecosystem can be structured into multiple interdependent layers, each with specific roles in sensing, communication, processing, and application. Fig. 1 provides a layered view of an IoE architecture that supports distributed intelligence and attack detection at multiple points in the system.

The base layer, known as the perception layer, comprises various IoT devices, including smart meters, thermostats, and energy consumption sensors. These devices collect operational data, but often have limited resources, making them more vulnerable to cyber threats [3]. Above the perception layer is the access layer, which is responsible for communication between the device layer and the edge layer. It uses protocols like ZigBee, Wi-Fi, 5G, powerline, and LoRaWAN to transmit data. On top of that is the edge AI layer, which uses lightweight models to analyze data near the source. This layer reduces latency and provides immediate responses in time-sensitive scenarios, such as voltage fluctuations or anomaly detection in substations [2]. Above the edge layer, the cloud layer acts as a central hub. It gathers information from numerous edge devices, performs more advanced analysis, checks for threats throughout the network, and helps enforce consistent policies [14], [15]. Finally, through the application layer, the user can interact with the system. This layer performs various functions and provides applications for real-time energy monitoring, dynamic

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pricing, and automatic billing. Humans can use this layer to enforce policies or verify alerts [11].

This layer structure prepares a flexible defence approach, where detection and basic analysis are performed on the edge layer, which is close to data sources, and more complex tasks are managed at the cloud layer. In this setup, the E-SHIELD system is designed to operate at the edge layer. It monitors real-time traffic from IoE devices to detect cyberattacks early and help prevent serious issues. By running the hybrid CNN-LSTM detection model directly on edge nodes, the E-SHIELD system quickly detects threats, such as DoS attacks, fuzzing, or malware intrusions, with minimal delay. It also uses SHAP for explainability, helping edge operators understand how the model reached its decision. This makes it easier to check alerts and respond on time. Because IoE edge devices have limited resources and need fast decisions, placing detection like E-SHIELD at this layer improves both speed and reliability of defence. Although centralized cloud-based analysis still plays a role in long-term storage and policy refinement, the primary goal of E-SHIELD is to provide lightweight, interpretable, and localized detection to protect critical energy infrastructure without relying on remote computation.

Methodology

The proposed E-SHIELD system, as shown in Fig. 2, integrates data processing, hybrid deep learning using CNN and LSTM layers, and a SHAP-based explainability module. The model extracts spatial-temporal patterns through a time-distributed convolutional layer followed by LSTM units and fully connected layers to perform binary classification. The final prediction is interpreted using SHAP, which allows network administrators to visualize the influence of input features on model decisions. This pipeline ensures high detection accuracy and transparency, supporting real-time intrusion monitoring in IoE networks.

Data Preprocessing and Feature Preparation

The network traffic from the WUSTL-IIoT-2021 and CICIoT-2023 datasets is preprocessed through cleaning and transformation step. Redundant and identifiable fields, such as source and destination addresses and port numbers, are removed to prevent model leakage. After this preprocessing, the final feature set contains 39 attributes, which are selected because they balance statistical, temporal, and protocol-level information for intrusion detection. Specifically, they include (i) statistical flow features such as packet counts, byte counts, rates, averages, and variances, (ii) temporal features such as inter-arrival times, jitter, and duration, and (iii) protocol-level indicators such as TCP, UDP, and ICMP flags. This combination allows the model to capture both bursty traffic behaviours and gradual attack dynamics while avoiding identifiers that could bias the model. Categorical variables are converted to numeric values using label encoding, and Z-score normalization is applied to scale continuous features. This improves convergence during training and

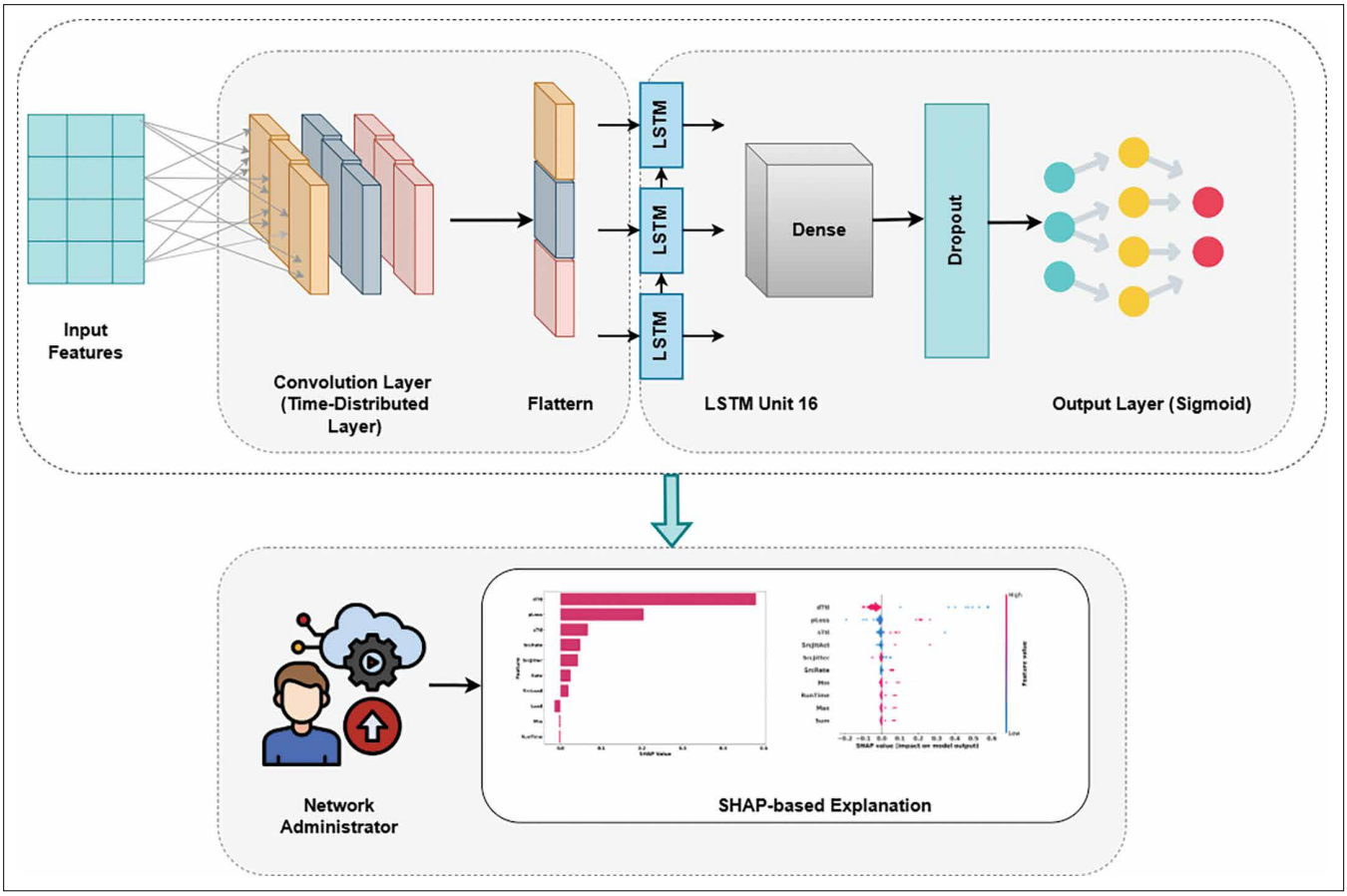


FIGURE 2. The E-SHIELD system architecture.

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ensures consistency across heterogeneous datasets. Since CICIoT-2023 and WUSTL-IIoT-2021 differ in data formats and labelling conventions, the preprocessing pipeline standardizes feature scales and unifies the attack into consistent categories (e.g., DoS, DDoS, fuzzing). To address class imbalance, the training set is balanced using the Synthetic Minority Oversampling Technique (SMOTE), which generates additional samples for underrepresented attack types, enabling the model to recognize rare patterns. Gaussian noise is added to the training data to improve robustness in real-world network variability. Finally, the data is split into training, validation, and testing sets using stratified sampling, ensuring that each subset maintains a proportional mix of attack types for accurate evaluation.

Attack Model

In this work, the adversary is assumed to launch network-based attacks such as DoS, DDoS, fuzzing, false data injection, and botnet variants, consistent with the attack types in the WUSTL-IIoT-2021 and CICIoT-2023 datasets. The attacker takes advantage of weak authentication or protocol flaws to overload resources, change sensor data, or disrupt normal operations. It is also assumed that the attacker does not physically tamper with IoE devices, alter training data, or introduce zero-day attacks.

Hybrid Deep Learning Architecture

The E-SHIELD system uses a hybrid CNN-LSTM model to detect unusual patterns in network traffic. The model is implemented using TensorFlow sequential API and contains the following main components:

1) Time-Distributed Conv1D Layer: The first layer in the model is a one-dimensional convolution layer that looks at the relationships between features at each point in time. It utilizes a small kernel size of 1, which enables it to focus on local patterns without introducing complexity. This layer is wrapped in a time-distributed wrapper, which means that the model examines each time step separately, thereby preserving the time order of the data. This setup is lightweight, with only four filters. It works well in limited-resource environments while still capturing important local interactions between traffic features. The ReLU activation adds non-linearity, enabling the model to detect irregular traffic behaviours without adding unnecessary computational cost.

2) Time-Distributed Flatten Layer: After the convolution step, the model uses a time-distributed flatten layer to reshape the data at each time step. This step reshapes the feature maps in a way that makes them ready for the next part of the model, where the LSTM layer learns patterns throughout the sequence.

3) LSTM Layer: A 16-unit Long Short-Term Memory (LSTM) layer processes the reshaped traffic data. It learns how behaviours change step by step over time. The LSTM learns sequential patterns of legitimate and malicious behaviour, helping the model to distinguish between fast and slow attacks. Unlike static classifiers, the recurrent nature of LSTM supports time-based analysis, which is vital in energy-based traffic streams where behaviour evolves across time windows.

4) Dense Layers and Dropout: The LSTM output is fed into a dense layer with 8 units and ReLU activation, followed by a dropout layer with a rate of 0.8. The high dropout rate ensures that the model does not overfit to dominant attack types and remains resilient to unseen patterns. The final dense layer has a single neuron with a sigmoid activation function that outputs a probability score indicating the probability of an attack.

The proposed hybrid deep learning architecture integrates 1D-CNN and LSTM to handle different types of IoE traffic. The 1D-CNN with a few filters provides an efficient way to capture sudden traffic bursts such as DoS or DDoS. The 16-unit LSTM layer models longer-term behaviours like false data injection or persistent botnet activity. Compared to GRU or standalone 1D-CNN, this combination of CNN for feature extraction and LSTM for sequential analysis offers stronger detection capability while remaining lightweight. The result is a compact but effective model that identifies various cyberattacks with minimal false positives.

Model Training and Performance Evaluation

To increase generalization, the model is trained using a five-fold stratified cross-validation technique. The training set is divided into five subsets so that each fold has the same distribution of classes. In each run, four folds are used for training and one fold for validation, and the process is repeated across all folds. This helps the model to train on different combinations of data and to generalize to new samples. A binary cross-entropy loss is calculated at each step and is optimized using Adam to achieve efficient convergence. Class weights are applied during training to increase the cost of misclassifying samples from the minority class, thus balancing the learning process. An early stopping mechanism monitors the validation loss, and training stops if there is no improvement over 30 epochs. This reduces overfitting and avoids unproductive training. The model evaluation is performed after cross-validation using the training and test sets. In practical IoE deployments, the model is updated through continual learning or periodic retraining to adapt to evolving traffic patterns and concept drift.

Explainable AI Interface and SHAP Integration

To support human-in-the-loop decision-making in critical infrastructure security, an interactive visual explanation interface is developed using the Streamlit framework. This interface is designed to assist cybersecurity analysts, grid operators, and network administrators by providing a transparent view of the predictions of the model and the underlying rationale for each decision. The key features of the interface include the following.

1) Prediction Viewer: For every selected input sample, the interface displays the predicted class label (e.g., attack or normal) with the associated probability score. A dynamic threshold slider allows users to explore the effects of varying the decision boundary, which can be useful to balance false positives and false negatives according to operational risk levels.

2) Local Explanations: SHAP values are generated using the Gradient Explainer for each sample to explain the contribution of individual

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features. These are displayed as intuitive horizontal bar plots that highlight the positive or negative influence of each feature on the prediction outcome. This functionality allows users to inspect the behaviour of the model on a case-by-case basis, which is necessary during forensic analysis or incident response.

3) Global Summary: A SHAP summary plot aggregates the feature importance across the entire dataset, helping users understand which input features consistently affect model decisions. This global explanation supports model auditing, risk assessment, and feature engineering for future model improvements.

4) Feature Viewer and Export: The interface includes options to view all feature values and SHAP explanations in tabular format. Users can download these outputs as CSV files for offline analysis, regulatory reporting, or compliance documentation, thus improving traceability and accountability in cybersecurity operations.

The explainability of SHAP can be integrated within the E-SHIELD system without changing the underlying hybrid DL system. This guarantees that performance benefits from CNN and LSTM layers are retained while further explainability is added. The E-SHIELD system incorporates not only trust, but also decision validation by providing a visualization of the prediction alongside its explanation. This makes it easier for operators to check if a decision is correct and take the proper steps to address potential threats in IoE networks.

Result Analysis

The E-SHIELD system is trained and tested using the WUSTL-IIoT-2021 [12] and CICIoT-2023 [13] datasets. These datasets include different real-world traffic and attack scenarios, which help DL models to learn how to detect various types of cyberattacks in IoE networks. The WUSTL-IIoT-2021 dataset is specifically designed for industrial and energy-based settings. It includes network traffic data from smart energy devices such as smart meters, Programmable Logic Controllers (PLCs), and other industrial sensors. These devices and sensors are part of the perception and control layers in IoE systems. This dataset contains both benign and malicious network traces of DoS, DDoS, Fuzzing, Mirai botnet, and Amnesia (AmY) attacks. These attacks consider IoE network attack scenarios in which attackers exploit device availability, manipulate sensor communication, or exploit protocol-level vulnerabilities. The CICIoT 2023 dataset includes a wide range of consumer IoT devices, such as smart cameras, voice assistants, smart thermostats, and home automation systems. This dataset complements WUSTL-IIoT-2021 by providing a wider range of device coverage. The greater number of devices reflects realistic traffic behaviours for edge nodes in residential or microgrid IoE networks. The dataset contains detailed labels for normal and attack traffic and includes modern DDoS threats like TCP SYN flood, UDP flood, HTTP flood, and Slowloris. These attacks target different layers of the protocol stack, from the transport layer to the application layer, making the dataset valuable for detecting multistage attacks in heterogeneous IoE environments. For example, SYN and UDP floods simulate resource exhaustion at gateways and edge routers, while HTTP-based attacks reflect overload attempts on web-based dashboards and cloud analytics platforms, which are common in energy monitoring applications. Both datasets are preprocessed before training. Features that reveal the identity of the source or destination, such as IP addresses and port numbers, are removed. Using the industrial relevance of WUSTL-IIoT-2021 with the broad device coverage of CICIoT-2023, the E-SHIELD system benefits from exposure to realistic, diverse, and well-labelled data.

Model performance is evaluated using accuracy, precision, recall, and the F1 score. During training, five-fold cross-validation is applied to ensure stable results. These metrics help to confirm that the models make reliable predictions and handle both normal and malicious traffic effectively. A user dashboard is designed to support explainability and improve understanding of the behaviour of the model. SHAP is used to calculate local and global feature contributions. On the user dashboard, the local explanation tab allows users to select a sample, view the prediction, and analyze the impact of the features. The global summary highlights the main features based on average SHAP values. All graphs and tables are interactive and can be exported. This setup allows network operators to trace predictions, review risk-related features, and provide informed responses during security incidents in energy-based IoT systems. The system is also compared with the state-of-the-art CNN [10], LSTM [9], and the DeepShield systems based on ensembles [6], showing improved accuracy across datasets.

Fig. 3 shows the accuracy results of different systems across the WUSTL-IIoT-2021 and CICIoT-2023 datasets. For the WUSTL-IIoT-2021 dataset, the E-SHIELD system records an accuracy of 0.96 with a stability of ±0.0059 under 95% confidence. Although the LSTM and DEEPShield systems also perform well, the E-SHIELD system shows a slight improvement. The LUCID system records the lowest accuracy in this group, showing that the CNN model may not fully capture

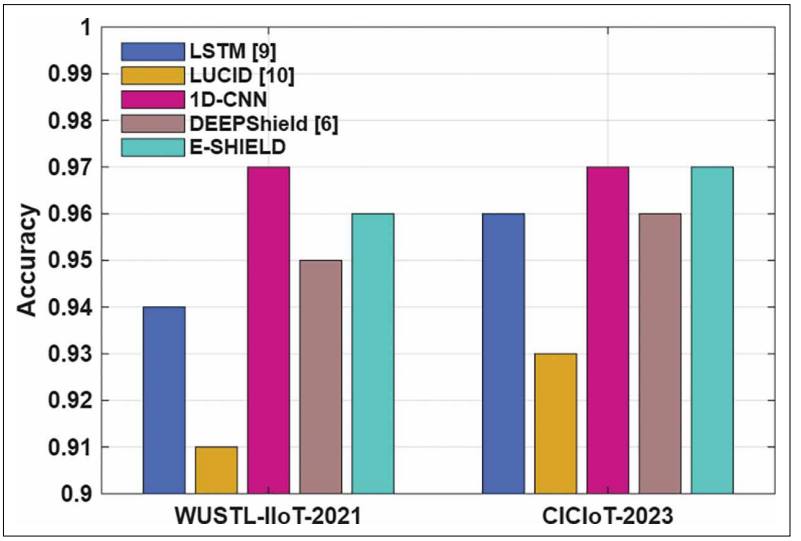


FIGURE 3. Systems Accuracy Comparison.

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temporal patterns in the data, while the 1D-CNN achieves an accuracy of 0.97. The results highlight the benefit of combining convolutional and sequential learning in a hybrid structure. On the CICIoT-2023 dataset, the E-SHIELD system again achieves an accuracy of 0.97, with a stability of ±0.005 under 95% confidence. This accuracy is comparable with the 1D-CNN. The E-Shield system performs slightly better than the DEEPShield and LSTM systems, which both achieve a score of 0.96. The LUCID system shows reduced accuracy compared to the others, which may indicate challenges in generalizing to complex attack patterns. The results show that the E-SHIELD system maintains strong performance across different datasets and system architectures.

Dataset	Model	Precision	Recall	F1 Score	Inference Time (s)	Memory Usage (Bytes)
WUSTL-IIoT-2021	LSTM [9]	0.93	0.93	0.93	0.0006	18765
	CNN (LUCID) [10]	0.89	0.90	0.89	0.0010	51609
	1D-CNN	0.90	0.96	0.93	0.0053	70633
	Ensemble CNN-LSTM (DEEPShield) [6]	0.90	0.91	0.90	0.0021	73000
	Hybrid CNN-LSTM (E-SHIELD)	**0.90**	**0.96**	**0.93**	**0.0013**	**19649**
CICIoT-2023	LSTM [9]	0.93	0.96	0.94	0.0009	21876
	CNN (LUCID) [10]	0.91	0.88	0.89	0.0012	77414
	1D-CNN	0.98	0.96	0.97	0.0034	89024
	Ensemble CNN-LSTM (DEEPShield) [6]	0.94	0.93	0.93	0.0028	78450
	Hybrid CNN-LSTM (E-SHIELD)	**0.97**	**0.96**	**0.96**	**0.0015**	**25805**

[1] https://github.com/Makhduma3/E-Shield

TABLE 1. Performance Comparison of E-SHIELD and Baseline Systems.[1]

Table 1 shows the performance of different DL systems on the WUSTL-IIoT-2021 and CICIoT-2023 datasets. The E-SHIELD system gives consistent results across both datasets. On WUSTL-IIoT-2021, the E-SHIELD system achieves a comparable F1 score (0.93) to the LSTM but improves recall by 3.2%. Compared to the LUCID system, it achieves a 4.5% higher F1 score and a 6.7% higher recall. Compared to the DEEPShield system, it offers a 3.3% recall gain with the same F1 Score. On CICIoT-2023, the E-SHIELD system achieves a precision of 0.97 and an F1 score of 0.96, which is 3.2% better than the DEEPShield system and about 9% better than the LUCID system. It also outperforms LSTM in precision by 4.3% while maintaining equal recall. All models are deployed on Raspberry Pi using ONNX, where

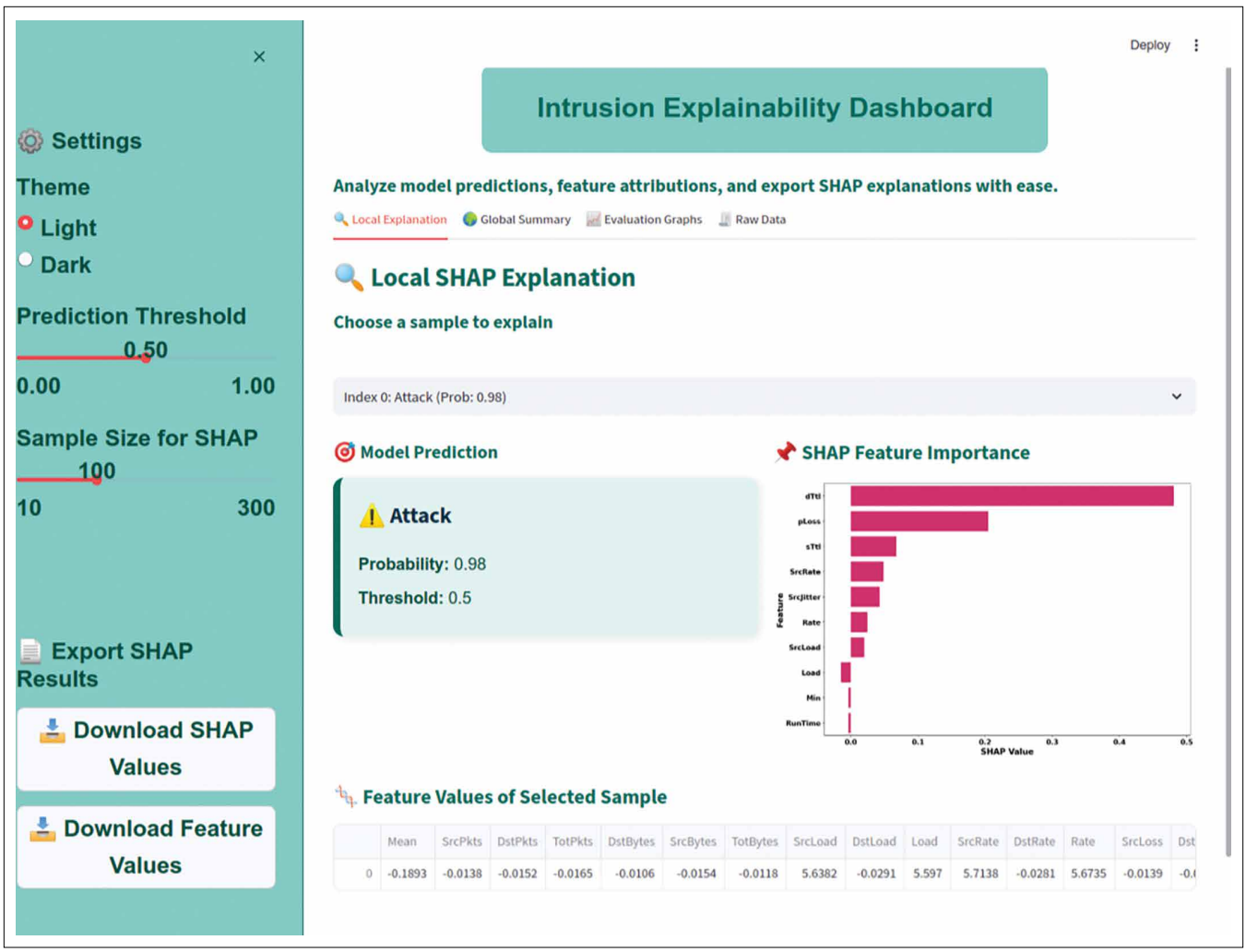


FIGURE 4. User Interface of E-SHIELD System.

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inference time and memory usage are measured to capture the performance–resource trade-off. For fairness, all measurements are obtained by passing the offline dataset samples in batches of 128. The E-SHIELD system performs well in this balance, running in 0.0013 seconds with 19.6 KB memory, which is 75% faster and 72% lighter than 1D-CNN (0.0053 seconds, 70.6 KB). Compared to LSTM, the E-SHIELD system provides the same recall but with a 30% lower inference time and greater stability across datasets. While the 1D-CNN model (32–16–16 filters) achieves strong accuracy (F1 = 0.97), its higher resource demands limit deployment. It also focuses only on local spatial patterns and, therefore, needs more filters, memory, and time.

The E-SHIELD system, with a 16-unit LSTM, 8-unit dense layer, and 0.8 dropout, is more compact, avoids overfitting, and captures temporal behaviour with lower resource use.

Fig. 4 shows the E-SHIELD explainable AI interface, designed to help analysts understand and verify model predictions. The dashboard provides a real-time local SHAP explanation for a selected network traffic sample, where the model predicts whether the instance is an attack or normal behaviour. The left sidebar allows users to adjust the prediction threshold and the number of samples used for SHAP computation. In the main panel, the predicted class label, probability score, and threshold are displayed, followed by a SHAP bar chart that highlights the most influential features for the decision. The interface also includes a detailed view of the feature values for the selected instance and options to export SHAP and feature data for offline analysis. By combining prediction outputs with interpretable visual explanations, the interface supports transparent decision-making and facilitates human-in-the-loop threat investigation in IoE networks. The user interface runs on a Digital Ocean droplet configured with one vCPU (DO-Regular at 2.3 GHz) and 2 GB RAM running Ubuntu, selected to approximate the resource constraints of edge devices. For a sample size of 100, SHAP generation records an average runtime of 1.23 seconds per explanation with a peak memory usage of 461.7 KB. The stability of explanations reaches a cosine similarity score of 0.948, and a regression-based surrogate gives a fidelity score of 0.999 R^2. For demonstration purposes, the interactive interface is deployed on the same droplet (http://206.189.207.55:8505/), allowing users to select sample sizes and observe these metrics directly. In addition, the modular design of the dashboard allows it to be deployed in large-scale IoE environments by distributing interface instances across nodes, and it can be linked with control systems to coordinate responses across different network segments. The dashboard can integrate with existing network or security centers by forwarding alerts and SHAP explanations. This helps operators match model output with network logs and speed up incident response.

Fig. 5 shows the SHAP summary plot for the WUSTL-IIoT-2021 dataset, generated using a sample of 50 instances. The plot highlights the main features that influence the predictions of the E-SHIELD system for cyberattack detection.

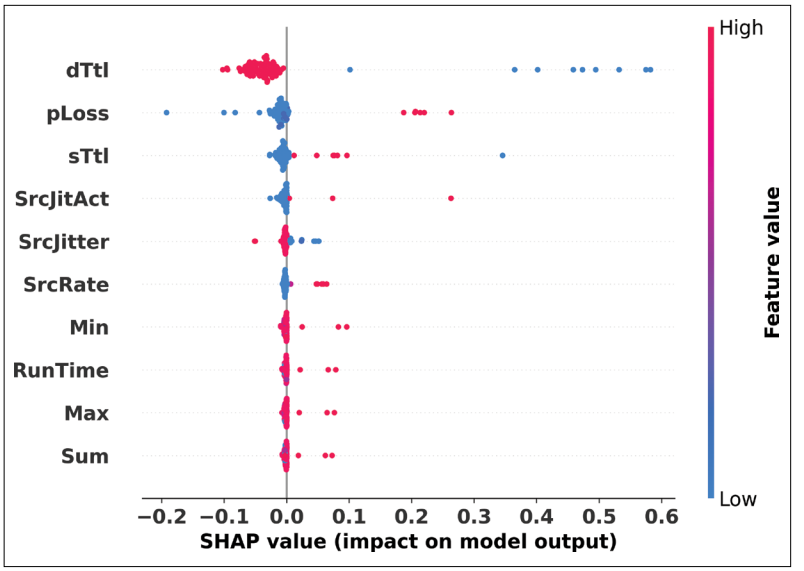


FIGURE 5. Feature Impact Analysis Using SHAP.

Each dot represents an SHAP value for a feature in a particular sample, with the color indicating the value of the feature (red for high and blue for low). The feature "dTtl" has the highest impact on the model prediction, where higher values are strongly related to the detection of attacks. Other important features include "pLoss" and "sTtl", which also influence the classification results. Features such as "SrcJitter" and "RunTime" contribute moderately, helping to detect variations in traffic behaviour.

Conclusion

This article introduced E-SHIELD, a novel explainable deep learning system for cyberattack detection in IoE networks. The system was built using a hybrid CNN-LSTM model. The model was trained and validated using five-fold stratified cross-validation to ensure robust and generalizable performance. Evaluation of the WUSTL-IIoT-2021 and CICIoT-2023 datasets showed that the E-SHIELD system outperformed existing systems in terms of precision, recall, and F1 score. Beyond detection, the system prioritized interpretability by incorporating SHAP for attribution and transparency of features. A fully interactive web-based dashboard that provides end users with access to local and global explanations, an exportable feature, and SHAP data. The interface enabled real-time decision support for network administrators and energy operators. By bridging the gap between high-performance detection and transparent decision making, the E-SHIELD system laid the foundation for future advancements in trustworthy, intelligent, and deployable cybersecurity solutions for energy systems. Future work will focus on strengthening the resilience of E-SHIELD against adversarial attacks and addressing concept drift in real-time energy systems. The work will also look at using federated learning to train IoE nodes in a privacy-preserving way. It will further expand the interface to support large-scale network deployment and integration with control systems.

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References

[1] M. Farhan et al., "Towards next generation Internet of Energy system: Framework and trends," *Energy AI*, vol. 14, Oct. 2023, Art. no. 100306.
[2] G. O. Alsharif, C. Anagnostopoulos, and A. K. Marnerides, "Energy market manipulation via false-data injection attacks," *IEEE Access*, vol. 13, pp. 42559–42573, 2025.
[3] M. Ghiasi et al., "Evolution of smart grids towards the Internet of energy: Concept and essential components for deep decarbonisation," *IET Smart Grid*, vol. 6, no. 1, pp. 86–102, Feb. 2023.
[4] D. Assante et al., "Internet of energy: New scenarios, opportunities, challenges and educational solutions," in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, Apr. 2021, pp. 1514–1520.
[5] M. U. Rehman and H. Bahşi, "Process-aware security monitoring in industrial control systems: A systematic review and future directions," *Int. J. Crit. Infrastruct. Protection*, vol. 47, Dec. 2024, Art. no. 100719.
[6] M. F. Saiyedand and I. Al-Anbagi, "Deep ensemble learning with pruning for DDoS attack detection in IoT networks," *IEEE Trans. Mach. Learn. Commun. Netw.*, vol. 2, pp. 596–616, 2024.
[7] S. H. Mohammed et al., "A review on the evaluation of feature selection using machine learning for cyber-attack detection in smart grid," *IEEE Access*, vol. 12, pp. 44023–44042, 2024.
[8] J. Zhang et al., "Improving multi-model anomaly traffic detection in MEC networks with large-model-powered continuous learning," *IEEE Netw.*, vol. 39, no. 3, pp. 56–62, May 2025.
[9] A. A. Alashhab et al., "Low-rate DDoS attack detection using deep learning for SDN-enabled IoT networks," *Int. J. Adv. Comput. Sci. Appl.*, vol. 13, no. 11, pp. 1–7, 2022.
[10] R. Doriguzzi-Corin et al., "Lucid: A practical, lightweight deep learning solution for DDoS attack detection," *IEEE Trans. Netw. Service Manage.*, vol. 17, no. 2, pp. 876–889, Jun. 2020.
[11] M. F. Saiyed, I. Al-Anbagi, and M. S. Hossain, "Interactive and explainable optimized learning for DDoS detection in consumer IoT networks," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 6839–6853, May 2025.
[12] M. Zolanvari et al., *Wustl-IIoT-2021 Dataset for IIoT Cybersecurity Research*. St. Louis, MI, USA: Washington Univ., 2021.
[13] E. C. P. Neto et al., "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, Jun. 2023.
[14] M. Al-Hawawreh and M. S. Hossain, "Federated learning-assisted distributed intrusion detection using mesh satellite nets for autonomous vehicle protection," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 854–862, Feb. 2024.
[15] M. G. Spina et al., "Distributing intelligence in 6G programmable data planes for effective in-network intrusion prevention," *IEEE Netw.*, vol. 39, no. 3, pp. 319–325, May 2025.

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